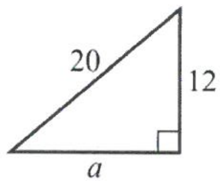
Lesson 1

Practice

3 In each of the following find the value of the pronumeral.

a

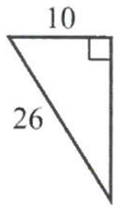


$20^2 - 12^2$

$= \sqrt{256}$

$= 16$

b

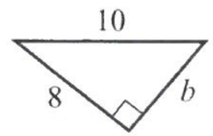


$26^2 - 10^2$

$a = \sqrt{576}$

$= 24$

c

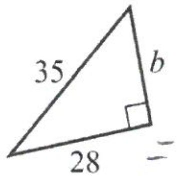


$10^2 - 8^2$

$= \sqrt{36}$

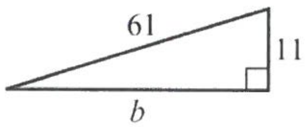
$= 6$

d



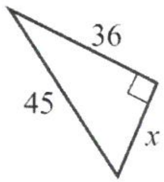
$= 21$

e



$= 60$

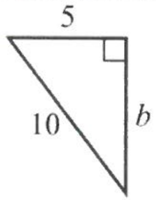
f



$= 27$

4 In each of the following, find the value of the pronumeral. Express your answers correct to <u>two decimal places.</u>

a

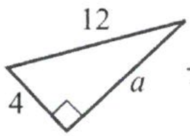


$10^2 - 5^2$

$= \sqrt{75}$

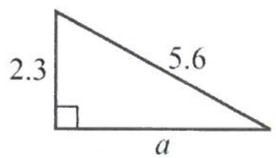
$= 8.66$

b



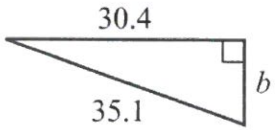
$= 11.31$

c



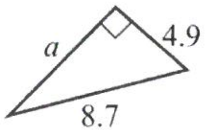
$= 5.11$

d



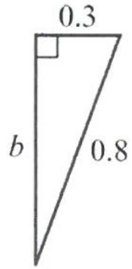
$= 17.55$

e



$= 7.19$

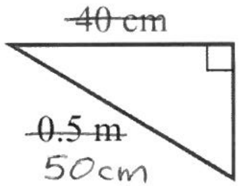
f

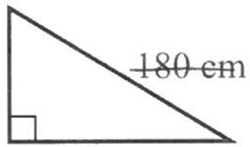


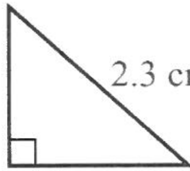
$= 0.74$

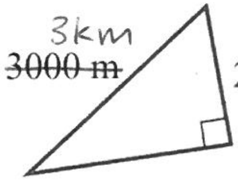
Extend

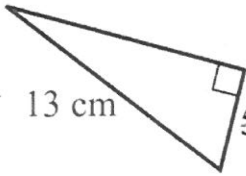
5 Find the length of the unknown side of each of these triangles, correct to two decimal places where necessary. Convert to the units shown in red.

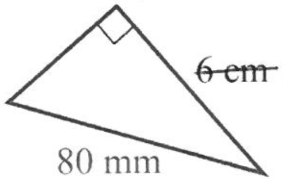
a


b


c


d


e


f


5a) 30cm

b) 149.67cm

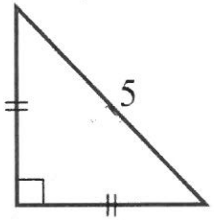
c) 1.65cm

d) 2.24 km

e) 12 cm

f) 52.92 mm

6 In each of the following, find the value of x as an exact answer.

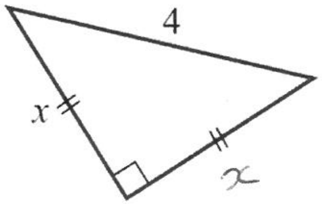
a


$x^2 + x^2 = 5^2$

$2x^2 = 25$

$x^2 = \frac{25}{2}$

$x = \sqrt{\frac{25}{2}} = \frac{5}{\sqrt{2}}$

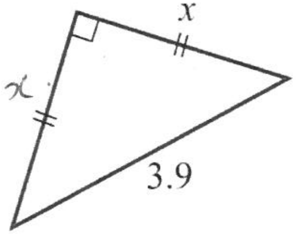
b


$x^2 + x^2 = 4^2$

$2x^2 = 16$

$x^2 = \frac{16}{2}$

$x = \sqrt{\frac{16}{2}} = \frac{4}{\sqrt{2}}$

c


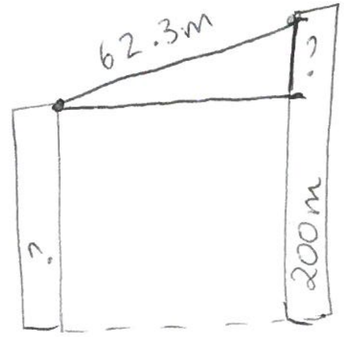
$= \frac{3.9}{\sqrt{2}}$

LESSON 3

lesson 3

You Do

2 Two skyscrapers are located 25 m apart and a cable of length 62.3 m links the tops of the two buildings. If the taller building is 200 metres tall, what is the height of the shorter building? Give your answer correct to one decimal place.



$62.3^2 - 25^2$

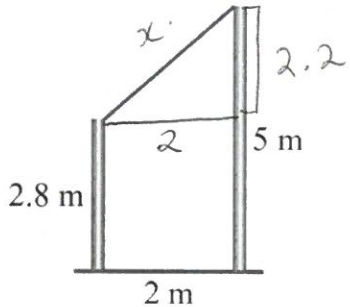
$= \sqrt{3256.29}$

$= 57.06m$

$200 - 57.06$

$= 142.9m$ tall.

3 Two poles are located 2 m apart. A wire links the tops of the two poles. Find the length of the wire if the poles are 2.8 m and 5 m in height. Give your answer correct to one decimal place.

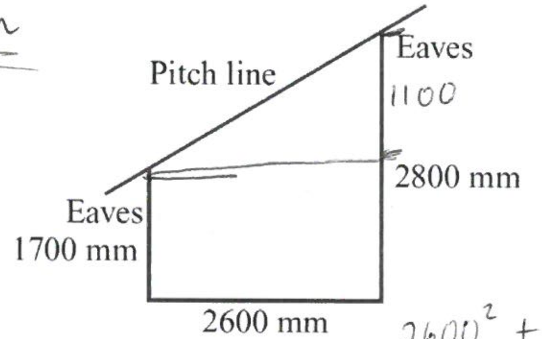


$= 2.2^2 + 2^2$

$= \sqrt{8.84}$

$= 2.97 \quad = 3.0 m$

4 A garage is to be built with a skillion roof (a roof with a single slope). The measurements are given in the diagram. Calculate the pitch line length, to the nearest millimetre. Allow 500 mm for each of the eaves.



$2600^2 + 1100^2$

$= \sqrt{7,970,000}$

$= 2,823.1mm$

+

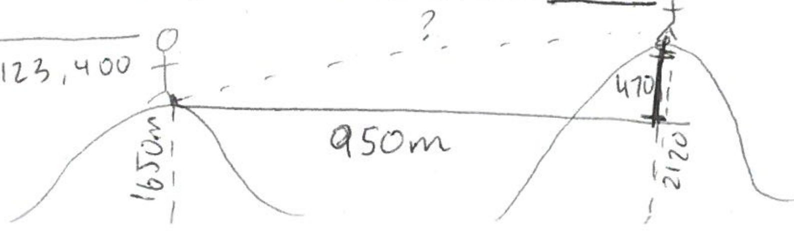
1000mm (eaves

$= 3823 mm$

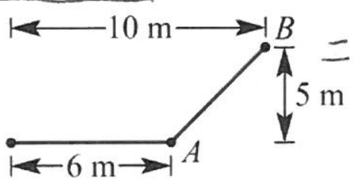
5 Two bushwalkers are standing on different mountain sides. According to their maps, one of them is at a height of 2120 m and the other is at a height of 1650 m. If the horizontal distance between them is 950 m, find the direct distance between the two bushwalkers. Give your answer correct to the nearest metre.

$950^2 + 470^2 = \sqrt{1,123,400}$

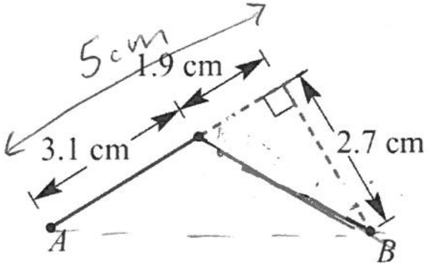
$= 1060m$



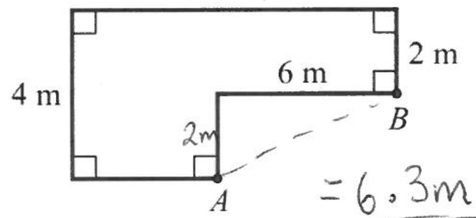
Extensio Find the direct distance between the points A and B in each of the following, correct to one decimal place.

a 

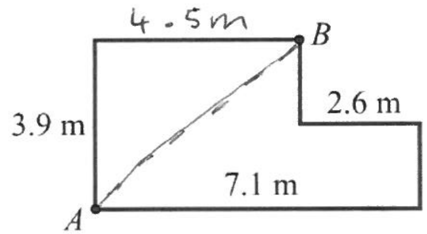
$= 6.4$m

b 

$= 5.7$cm

c 

$= 6.3$m

d 

$= 5.95$

$= 6.0$m

7 A 100 m radio mast is supported by six cables in two sets of three cables. They are anchored to the ground at an equal distance from the mast. The top set of three cables is attached at a point 20 m below the top of the mast. Each cable in the lower set of three cables is 60 m long and is attached at a height of 30 m above the ground. If all the cables have to be replaced, find the total length of cable required. Give your answer correct to two decimal places.